

On a moment determinacy conjecture of Bertoin and Yor

Martin Minchev*

Abstract

Let ξ be an unkilled real-valued Lévy process which drifts to $+\infty$ and has positive exponential moments of all orders, and define

$$I_\xi = \int_0^\infty e^{-\xi_t} dt, \quad \text{and its reciprocal } X_\xi = 1/I_\xi.$$

Bertoin and Yor [BY02] proved that X_ξ is moment-determinate when ξ has no positive jumps, and conjectured that this condition is also necessary. We prove the latter.

The proof is based on a lower bound near zero for the law of I_ξ . We show that a group of sufficiently many positive jumps near the origin puts I_ξ on a suitable small scale. The first selected jump time is used as a one-dimensional smooth coordinate, yielding an absolutely continuous subcomponent of the law of I_ξ . After the change of variables, the resulting subdensity of X_ξ satisfies a Krein moment indeterminacy criterion.

Keywords. Exponential functionals; Lévy processes; moment problem; Krein criterion.

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1 Introduction

Let ξ be an unkilled Lévy process with Laplace exponent

$$\psi(q) = \log \mathbb{E}(e^{q\xi_1}), \quad \text{for } q \geq 0,$$

and suppose throughout that

$$\psi(q) < \infty \quad \text{for all } q \geq 0, \quad 0 < \psi'(0+) = \mathbb{E}(\xi_1) < \infty. \quad (1)$$

Then ξ drifts to $+\infty$, and the exponential functional

$$I_\xi = \int_0^\infty e^{-\xi_t} dt \quad \text{is finite a.s.}$$

Bertoin and Yor [BY02] proved that its reciprocal has all moments given by

$$\mathbb{E}(I_\xi^{-k}) = \psi'(0+) \frac{\psi(1)\psi(2) \cdots \psi(k-1)}{(k-1)!}, \quad \text{for } k \geq 1, \quad (2)$$

and showed that, when ξ has no positive jumps, $1/I_\xi$ has an exponential moment in a neighbourhood of the origin and is therefore moment-determinate. They conjectured that the absence of positive jumps is also necessary; see also [BY05, MS26].

The aim of this note is to prove this necessity. If positive jumps are present, then $\psi(q)$ grows at least exponentially along large q , so classical Carleman-type criteria fail; for background see, for example, [Lin17, SLK20]. However, this does not imply moment indeterminacy on its own,

*Institute of Mathematics, University of Zurich, Switzerland. Faculty of Mathematics and Informatics, Sofia University, Bulgaria. Email: martin.minchev@math.uzh.ch.

and one needs additional information on the density of I_ξ near zero. One possible approach, as described in [MS26, Remark 3.7], would be to use Mellin inversion and saddle-point analysis of the Bernstein–gamma functions introduced by Patie and Savov [PS18], in the spirit of the density asymptotics obtained in [MS23]. Here we obtain the required lower bound probabilistically. In the pure Poisson case, Bertoin and Yor [BY02, Section 4] obtained explicit small-value density information for I_ξ . After inversion, this gives a lognormal-type tail for $1/I_\xi$, in the classical scale of moment indeterminacy. Here we prove a lower bound of the same logarithmic order.

Consequently, a subdensity version of the Krein criterion, Proposition 2.1, provides the main result.

Theorem 1.1. *Assume (1). Then*

$$1/I_\xi \text{ is moment-determinate} \iff \Pi((0, \infty)) = 0,$$

where Π denotes the Lévy measure of ξ .

Only the forward direction has not been proved. The argument can be summarised as follows: if m positive jumps of size at least a occur near time zero, then the future part of the exponential functional is multiplied by roughly e^{-am} . Thus a natural scale for small values of I_ξ is

$$\ell_m = Le^{-am}, \quad \text{for some } L > 0.$$

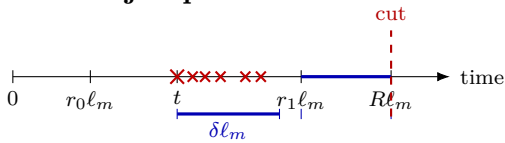
The number m of jumps makes the future contribution small. The first selected jump time t is then varied in an interval of order ℓ_m . Before this first selected jump, the selected compound Poisson part has not moved, so the contribution to I_ξ is of order t and *a fortiori* ℓ_m as well. By choosing suitably endpoints of the interval in which t changes, the monotone map $t \mapsto I(t)$ crosses the whole

$$\mathcal{I}_m = (\ell_m, \ell_{m-1}].$$

Since the first selected jump time has a density and this map has a uniform one-sided Lipschitz bound on the controlled event, this gives an absolutely continuous subcomponent of the law of I_ξ on each $\mathcal{I}_m$. Thus we obtain a density lower bound.

The mechanism is sketched in the figure below. The constants r_0, r_1, δ and R are fixed later in the proof; here the figure only indicates their roles.

(a) selected jumps



(b) density mechanism

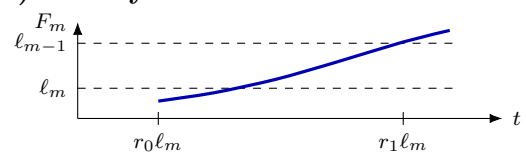


Figure 1: The group of jumps construction. Left: the first selected jump time T is allowed to take values $t \in [r_0\ell_m, r_1\ell_m]$, and the remaining selected jumps occur within a time interval of length $\delta\ell_m$ after it, before the deterministic cut time $R\ell_m$. Right: after the remaining randomness is frozen, the map $t \mapsto F_m(t)$ is increasing and crosses the scale interval $\mathcal{I}_m = (\ell_m, \ell_{m-1}]$.

1.1 Use of AI tools

OpenAI’s ChatGPT was used during the exploratory and editorial stages of this work, including GPT-5.4 Thinking and GPT-5.5 variants such as Thinking and Pro. In particular, the idea of trying a direct construction based on an early group of positive jumps, and then varying the time of the first selected jump as a smooth coordinate, came up in a *discussion* with it.

The use of a selected jump time can be seen in Bertoin, Lindner, and Maller [BLM08, Section 3.3]: they condition on almost all random quantities except certain jump times, so that the remaining conditional law is obtained as the image of Lebesgue measure under a deterministic map. The author takes full responsibility for the manuscript.

2 A Krein criterion for subdensities

We first record the corollary of the usual Krein criterion which we use later. If μ is a probability measure, we call a finite positive measure ν a *submeasure* of μ if $\nu(A) \leq \mu(A)$ for every measurable set A .

Proposition 2.1 (Krein criterion for subdensities). *Let μ be a probability measure on $(0, \infty)$ with moments of all positive orders. Suppose that there exists a non-zero absolutely continuous submeasure*

$$\nu(dx) = h(x) dx \leq \mu(dx)$$

such that ν has moments of all positive orders and

$$\int_c^\infty \frac{-\log h(x^2)}{1+x^2} dx < \infty, \quad \text{for some } c > 0, \quad (3)$$

where h is positive on (c^2, ∞) . Then μ is moment-indeterminate.

Proof. Define the probability measure $\tilde{\nu}(dx) = \nu(dx)/\nu((0, \infty))$. Multiplying the density by a positive constant does not affect the finiteness of the logarithmic integral. Hence the half-line form of Krein's criterion [Lin17, Theorem 4] applies to $\tilde{\nu}$. Thus there exists a probability measure $\tilde{\nu}_\bullet \neq \tilde{\nu}$ on $(0, \infty)$ such that

$$\int_0^\infty x^n \tilde{\nu}(dx) = \int_0^\infty x^n \tilde{\nu}_\bullet(dx), \quad \text{for } n \geq 0.$$

Writing $\mu = \nu + \mu_0$, with μ_0 a positive finite measure, define $\tilde{\mu} = \nu((0, \infty)) \tilde{\nu}_\bullet + \mu_0$. Then $\tilde{\mu}$ is a probability measure whose moments coincide with those of μ , while $\tilde{\mu} \neq \mu$. Therefore μ is moment-indeterminate. $\square$

3 Density bound in the positive-jump case

Assume in this subsection that

$$\Pi((0, \infty)) > 0, \quad \text{and choose } a > 0 \text{ such that } \lambda = \Pi([a, \infty)) > 0.$$

Writing $\Delta\xi_s = \xi_s - \xi_{s-}$ for the jump of ξ at time s , set

$$S_t = \sum_{s \leq t} \Delta\xi_s \mathbf{1}_{\{\Delta\xi_s \geq a\}}, \quad \text{and} \quad \eta_t = \xi_t - S_t.$$

Then S is a compound Poisson process of rate λ , independent of the Lévy process η .

Proposition 3.1. *There exist constants $c_1, c_2 > 0$ and an absolutely continuous submeasure*

$$\nu_{I_\xi}(dy) = h_{I_\xi}(y) dy \leq \mathbb{P}(I_\xi \in dy)$$

such that

$$h_{I_\xi}(y) \geq c_1 \exp(-c_2(\log(1/y))^2), \quad \text{for all sufficiently small } y > 0. \quad (4)$$

Proof. For lighter notation, we write I instead of I_ξ in the proof. Fix $K > 0$. Since η is a càdlàg Lévy process starting from 0, we can choose $A_0 > 0$ such that

$$p_0 = \mathbb{P}\left(\sup_{0 \leq s \leq A_0} |\eta_s| \leq K\right) > 0. \quad (5)$$

Choose $r_0, r_1, \delta > 0$ such that

$$e^K(r_0 + \delta) < 1/4, \quad e^{-K}r_1 > e^a, \quad \text{and} \quad R = r_1 + \delta. \quad (6)$$

This choice will be used only for the crossing estimate

$$F_m(r_0\ell_m) < \ell_m, \quad F_m(r_1\ell_m) > \ell_{m-1},$$

sketched in the right panel of Figure 1. Next, choose $M > 0$ such that $\mathbf{P}(I \leq M) > 0$. For a positive constant L , which we choose shortly, and $m \geq 1$, we introduce the natural scale

$$\ell_m = Le^{-am}, \quad \mathcal{I}_m = (\ell_m, \ell_{m-1}]. \quad (7)$$

At the deterministic cut time $R\ell_m$, the Markov property gives

$$I = \int_0^{R\ell_m} e^{-\xi_s} ds + e^{-\xi_{R\ell_m}} I', \quad (8)$$

where I' is an independent copy of I . We choose L so that, on $\{I' \leq M\}$, the second term in (8) is negligible on the scale ℓ_m after the m selected jumps. Indeed, on the controlled event, after these jumps have occurred,

$$e^{-\xi_{R\ell_m}} I' \leq e^K e^{-am} M = \frac{e^K M}{L} \ell_m, \quad \text{thus choose } L > 0 \text{ so that } e^K M \leq \frac{L}{8}. \quad (9)$$

Fix m large enough so that $R\ell_m \leq A_0$, so we can use (5). We construct a submeasure of $\mathcal{L}(I)$ with density bounded from below on $\mathcal{I}_m$. Leave the first selected jump time as a continuous variable

$$t \in [r_0\ell_m, r_1\ell_m],$$

and impose the following selected-jump pattern:

- the first jump of S occurs at time t ;
- exactly $m - 1$ further jumps of S occur in $(t, t + \delta\ell_m]$;
- no other jump of S occurs before $R\ell_m$;
- $\sup_{0 \leq s \leq R\ell_m} |\eta_s| \leq K$;
- in the Markov decomposition at time $R\ell_m$, the independent copy I' satisfies $I' \leq M$.

The fourth condition has probability at least p_0 by (5), and the fifth has probability $\mathbf{P}(I \leq M) > 0$.

Let T be the first jump time of S , and let $N_S(B)$ denote the number of jumps of S in a Borel set B . For every Borel set $B \subseteq [r_0\ell_m, r_1\ell_m]$, the selected-jump conditions give

$$\begin{aligned} \mathbf{P}(T \in B, N_S((T, T + \delta\ell_m]) = m - 1, N_S((T + \delta\ell_m, R\ell_m]) = 0) \\ = \lambda e^{-\lambda R\ell_m} \frac{(\lambda \delta\ell_m)^{m-1}}{(m-1)!} \text{Leb}(B). \end{aligned} \quad (10)$$

Indeed, conditionally on $T = t$, the increments of S after time t are independent of the past, and the conditional probability of the two count conditions equals

$$e^{-\lambda \delta\ell_m} \frac{(\lambda \delta\ell_m)^{m-1}}{(m-1)!} e^{-\lambda(R\ell_m - t - \delta\ell_m)}.$$

Multiplying by the density $\lambda e^{-\lambda t}$ of T gives (10).

Freeze the path of η on $[0, R\ell_m]$, the selected jump sizes, the relative positions of the $m - 1$ jumps in $(t, t + \delta\ell_m]$, and the value of I' , under the two control conditions above. Let $F_m(t)$ be the value of I obtained from this frozen configuration. We now check the crossing property, see the right panel of Figure 1, namely

$$F_m(r_0\ell_m) < \ell_m, \quad \text{and} \quad F_m(r_1\ell_m) > \ell_{m-1}.$$

At $t = r_0\ell_m$, the contribution before and during the group is at most $e^K(r_0 + \delta)\ell_m$. After the group, all m selected jumps have occurred, so the remaining contribution, including the second term in (8), is at most

$$e^K e^{-am}(R\ell_m + M).$$

By (6) and taking m large enough,

$$F_m(r_0\ell_m) < \ell_m.$$

At the other endpoint, before the first selected jump the selected compound Poisson part has not moved. Hence

$$F_m(r_1\ell_m) \geq \int_0^{r_1\ell_m} e^{-\eta_s} ds \geq e^{-K} r_1\ell_m > e^a \ell_m = \ell_{m-1}.$$

Thus, once continuity is established, the image of $[r_0\ell_m, r_1\ell_m]$ under F_m contains $\mathcal{I}_m$.

The map F_m is increasing, since delaying the selected group can only increase $e^{-\xi_s}$ pointwise. Moreover, if the first selected jump time is shifted from t to $t + u$, then each of the m selected jump times is shifted by u . Thus the two integrands can differ only on at most m time intervals of length u . On the controlled event, $e^{-\xi_s} \leq e^K$, and the terminal term in (8) is unchanged, since all selected jumps have occurred before $R\ell_m$ in both configurations. Therefore

$$0 \leq F_m(t + u) - F_m(t) \leq e^K m u, \quad (11)$$

whenever $t, t + u \in [r_0\ell_m, r_1\ell_m]$. In particular F_m is continuous. Since its image contains $\mathcal{I}_m$, (11) implies that, for every Borel set $A \subset \mathcal{I}_m$,

$$\text{Leb}\{t \in [r_0\ell_m, r_1\ell_m] : F_m(t) \in A\} \geq e^{-K} m^{-1} \text{Leb}(A). \quad (12)$$

Now apply (10) with

$$B = \{t \in [r_0\ell_m, r_1\ell_m] : F_m(t) \in A\}.$$

Let $A \subset \mathcal{I}_m$ be Borel. The estimates above are uniform in the frozen variables satisfying the two control conditions. Hence the contribution to $\mathbf{P}(I \in A)$ coming from the selected-jump construction is bounded below by

$$p_0 \mathbf{P}(I \leq M) \lambda e^{-\lambda R\ell_m} \frac{(\lambda \delta \ell_m)^{m-1}}{(m-1)!} e^{-K} m^{-1} \text{Leb}(A).$$

Here p_0 and $\mathbf{P}(I \leq M)$ come from the two control conditions, (10) gives the selected-jump contribution, and (12) gives the lower bound on the set of first-jump times leading to $I \in A$. The jump sizes and the relative positions of the $m - 1$ later jumps are then integrated over their conditional laws.

Since $\ell_m \rightarrow 0$, the factor $e^{-\lambda R\ell_m}$ is bounded from below for all large m . Absorbing all fixed positive constants into $c > 0$, we get

$$\mathbf{P}(I \in A) \geq c m^{-1} \frac{(c\ell_m)^{m-1}}{(m-1)!} \text{Leb}(A), \quad \text{for all Borel } A \subset \mathcal{I}_m.$$

Equivalently, on $\mathcal{I}_m$ the law of I dominates an absolutely continuous measure with density h_m satisfying

$$h_m(y) \geq c m^{-1} \frac{(c\ell_m)^{m-1}}{(m-1)!}, \quad \text{for } y \in \mathcal{I}_m, \quad (13)$$

where $c > 0$ is independent of m and y . Consider then

$$h_I(y) = \sum_{m \geq m_0} \mathbf{1}_{\mathcal{I}_m}(y) h_m(y),$$

where m_0 is large enough for the preceding estimates to hold. Since the intervals $\mathcal{I}_m$ are disjoint and each $h_m(y) dy$ is dominated by $\mathcal{L}(I)$ on $\mathcal{I}_m$, the measure $h_I(y) dy$ is also dominated by $\mathcal{L}(I)$.

It remains to estimate (13). By Stirling's formula and (7),

$$-\log \left(m^{-1} \frac{(c\ell_m)^{m-1}}{(m-1)!} \right) \leq C_1 m^2, \quad \text{for large } m.$$

Moreover, for $y \in \mathcal{I}_m = (\ell_m, \ell_{m-1}]$ with $\ell_m = Le^{-am}$, we have that

$$a(m-1) - \log L \leq \log(1/y) < am - \log L.$$

Thus m is comparable with $\log(1/y)$ uniformly for $y \in \mathcal{I}_m$. Consequently, there exist constants $c_1, c_2 > 0$ such that

$$h_I(y) \geq c_1 \exp(-c_2(\log(1/y))^2), \quad \text{for all sufficiently small } y > 0.$$

□

4 Proof of the main result

Theorem 1.1 is now a straightforward consequence of the previous propositions.

Proof of Theorem 1.1. Assume first that $\Pi((0, \infty)) > 0$, and let h_{I_ξ} be the subdensity from Proposition 3.1. The change of variables $x = 1/y$ gives an absolutely continuous submeasure of the law of $X_\xi = 1/I_\xi$ with density

$$h_{X_\xi}(x) = x^{-2} h_{I_\xi}(1/x).$$

From (4),

$$h_{X_\xi}(x) \geq c_1 x^{-2} \exp(-c_2(\log x)^2), \quad \text{for large } x.$$

Consequently, for some $C_1, C_2, C_3 \geq 0$,

$$-\log h_{X_\xi}(x^2) \leq C_1 + C_2 \log x + C_3(\log x)^2, \quad \text{for large } x.$$

It follows that, for a suitable $c > 0$,

$$\int_c^\infty \frac{-\log h_{X_\xi}(x^2)}{1+x^2} dx < \infty.$$

Therefore Proposition 2.1 implies that X_ξ is moment-indeterminate.

For completeness, we recall the argument for the reverse direction, as in [BY02]. If $\Pi((0, \infty)) = 0$, then $\psi(q) = O(q^2)$ as $q \rightarrow \infty$ by the Lévy–Khintchine formula. Hence (2) gives

$$\mathbb{E}(X_\xi^k) = \mathbb{E}(I_\xi^{-k}) \leq C^k k!, \quad \text{for all } k \geq 1.$$

Thus $\mathbb{E}(e^{\theta X_\xi}) < \infty$ for some $\theta > 0$, and therefore X_ξ is moment determinate. □

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